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CAT WAND TOY

Abstract

A pet toy includes a handle, at least one light source, a transparent or semitransparent shaft extending from the handle including a plurality of optical strands, a cord attached to the shaft, and a toy assembly coupled to the cord. The one or more optical strands are configured to radiate light generated by the at least one light source.

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Background/Summary

[0001] This application claims the benefit of priority to U.S. Provisional Application entitled “LED Cat Wand Toy,” filed Mar. 10, 2024, and assigned Ser. No. 63/563,390, the entire disclosure of which is hereby expressly incorporated by reference.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The disclosure relates generally to pet toys, and more specifically, to interactive pet toys including multiple functionalities configured to mentally stimulate pets.

BRIEF DESCRIPTION OF RELATED TECHNOLOGY

[0003] Generally, a wide variety of pet toys are available to keep pets entertained. For example, pet toys including one or more of a squeaker, a bell, crinkle paper, and the like may be configured to make noise, entertaining pets. Other pet toys include balls, frisbees, wands, toys including treats (e.g., cat nip, pet treats), chew toys, and the like. However, pets may quickly become disinterested in their toys. Accordingly, there is a need for more dynamic and interactive pet toys including multiple functionalities that engage pets for longer or extended periods of time.

SUMMARY OF THE DISCLOSURE

[0004] In accordance with one aspect of the present disclosure, a pet toy includes a handle, at least one light source, a transparent or semitransparent shaft extending from the handle including a plurality of optical strands, a cord attached to the shaft, and a toy assembly coupled to the cord. The one or more optical strands are configured to radiate light generated by the at least one light source.

[0005] In accordance with another aspect of the present disclosure, a pet toy includes a handle, at least one light source, a transparent shaft extending from the handle, the shaft including one or more optical strands, a cord attached to the shaft, a toy assembly coupled to the cord, the toy assembly including at least one additional light source. The one or more optical strands are configured to radiate light generated by the at least one light source.

[0006] In accordance with yet another aspect of the present disclosure, a pet toy includes a handle including at least one user input device, at least one light source, a transparent shaft extending from the handle, the shaft including one or more optical strands, a cord attached to a distal end of the shaft opposite the handle.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0007] For a more complete understanding of the disclosure, reference should be made to the following detailed description and accompanying drawing figures, in which like reference numerals identify like elements in the figures.

[0008] FIG. **1** illustrates a front view of a pet toy in accordance with one example of the present disclosure.

[0009] FIG. **2** illustrates a side view of a handle of the pet toy of FIG. **1** in accordance with one example of the present disclosure.

[0010] FIG. **3** illustrates a partial cross section view of a pet toy in accordance with one example of the present disclosure.

[0011] FIG. **4** illustrates an optical strand in accordance with one example of the present disclosure.

[0012] FIG. **5** illustrates a block diagram of a pet toy in accordance with one example of the present disclosure.

[0013] FIG. **6** illustrates a partial back view of a pet toy in accordance with one example of the present disclosure.

[0014] FIG. **7** illustrates a partial front view of pet toy in accordance with one example of the present disclosure.

[0015] While the disclosed pet toys and methods are susceptible of embodiments in various forms, there are illustrated in the drawings (and will hereafter be described) specific embodiments of the disclosure, with the understanding that the disclosure is intended to be illustrative and is not intended to limit the disclosure to the specific embodiments described and illustrated herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0016] The present disclosure is provided to solve the above-mentioned problems in the Background of the Disclosure section. Specifically, the present disclosure provides a pet toy having multiple functionalities providing a unique and dynamic playing experience for pets. In accordance with the present disclosure, a pet toy may include a handle, a clear (e.g., transparent, semi-transparent) shaft, one or more optical strands disposed in the shaft, at least one light source, a cord attached to the shaft, and a toy assembly removably coupled to the chord. According to the present disclosure, the one or more optical strands may be configured to radiate light generated by the at least one light source. The light radiated by the one or more optical strands may be visible through the shaft so as to engage or capture the attention of a pet.

[0017] In accordance with the present disclosure, when a person holds the handle, the toy assembly may hang or dangle, via the cord, from the shaft of the pet toy. Accordingly, a person holding the handle may dangle the toy assembly in front of a pet, enticing the pet to bat at or otherwise play with the toy assembly. In some examples, the toy assembly may include an attractant coupled to a body of the toy assembly configured to further attract or capture the attention of a pet. In addition to the toy assembly, the one or more optical strands may radiate light to attract or capture the attention of a pet. According to some examples, the one or more optical strands may be disposed in the shaft so as to form a recurring pattern or shape. According to some examples, the one or more optical strands may radiate light according to two or more different operational modes. In each of the operational modes, the one or more optical strands may radiate light having different colors (i.e., an assortment of colors), having different intensities, and/or in different sequences in which the one or more optical strands alternate between lit and unlit conditions, change colors, and/or change brightness.

[0018] According to some examples, the toy assembly may include an additional light source configured generate light for attracting or capturing the attention of a pet. In accordance with some examples, light may be alternating radiated by the one or more optical strands and the toy assembly so as to move a pet's attention between the one or more optical strands and the toy assembly. According to some examples, as described below in greater detail, a toy assembly according to the present disclosure may include one or more of catnip, bright colors, distinct or contrasting colors or images, distinct textures, and/or contrasting textures to attract or capture the attention of a pet.

[0019] Referring to FIG. **1**, a front view of a cat wand or pet toy **100** is illustrated in accordance with one example of the present disclosure. As shown in FIG. **1**, the pet toy **100** may include a handle **110**, a shaft **120** extending from the handle **110**, one or more optical strands **130** disposed within the shaft **120**, a light source **140**, a cord **150** attached to the shaft **120**, and a toy assembly **160** coupled to the cord **150**. According to other examples, additional, fewer, or different components may be included.

[0020] The handle **110** may be configured to be held by a person or human playing with a pet. The handle **110** may have an ergonomic shape so as to be comfortably held within (e.g., a single) hand of a human. As noted above the toy assembly **160** may hang or dangle, via the cord **150**, from the shaft **120** of the pet toy **100** when a person holds the handle **110** of the pet toy **100**. In some examples, the handle **110** may be comprised of a plastic or plastics, such as, polypropylene (PP), polyethylene, polycarbonate, or other similar materials. In other examples, the handle **110** may be comprised of other materials.

[0021] Referring to FIG. **2**, a side view of a handle **110** of the pet toy **100** is illustrated in accordance with one example of the present disclosure. In some examples, as illustrated in FIG. **2**, the handle **110** may include a front or first panel **111** and a back or second panel **112**. In these examples, the first panel **111** and the second panel **112** may be coupled to one another, forming the handle **110**. One or more fasteners, for example, screws, bolts, pins, rivets, nails, and the like may be used to couple the first panel **111** and the second panel **112**.

[0022] Returning to FIG. **1**, the pet toy **100** may further include a shaft **120** extending from the handle **110** (e.g., extending from one end of the handle **110**). In some examples, the shaft **120** may be formed separately from and subsequently coupled to the handle **110**. For example, the shaft **120** may be disposed between the first panel **111** and the second panel **112** of the handle **110**. In some examples, the shaft **120** and the first panel **111** and/or second panel **112** may include mating features configured to maintain a position of the handle **110** and shaft **120** relative to one another. For example, the shaft **120** may include a rib or protrusion configured to be received in a groove or receptacle in the first panel **111** and/or second panel **112** of the handle **110** (e.g., disposed at an inner surface of the first panel **111** and/or second panel **112**). In other examples, the shaft **120** and the handle **110** may be integrally formed as a single element of the pet toy **100**.
[0023] According to some examples, as shown in FIG. **1**, the shaft **120** may have a circular cross-sectional shape; however, the present disclosure is not limited thereto. In other examples, the shaft **120** may have a polygonal, elliptical, oval, or another cross-sectional shape. The shaft **120** may be clear (e.g., transparent, semi-transparent), such that light radiated by the one or more optical strands **130** is visible through the shaft **120**. In some examples, the one or more optical strands **130** may be visible through the shaft **120**. In some examples, the shaft **120** may be made of a plastic or plastics, such as acrylic, for example, polymethyl methacrylate (PMMA), or other similar materials. In accordance with other examples, the shaft **120** may be made of polypropylene (PP), polyethylene, polycarbonate, or other similar materials. In other examples, the shaft **120** may be made of another material. The shaft **120** includes a proximal end **121** disposed at or adjacent to the handle **110** and a distal end **122** opposite the proximal end **121**. The distal end **122** is disposed further away from the handle **110** than the proximal end **121**.
[0024] Referring generally to FIGS. **1** and **2**, the handle **110** may further include one or more user input devices **113**, for example, a button, switch, knob, lever, dial, touch or proximity sensor, or the like. The one or more user input devices **113** may be operatively coupled to the at least one light source **140**. For example, the one or more user input devices **113** may be configured to turn the at least one light source **140** on and/off, change a functionality or operating mode of the at least one light source **140**, and the like. An ergonomic shape of the handle **110** may further allow a person to operate one or more user input devices **113**, for example, a button or switch, included in the handle **110**, while comfortably holding the handle **110** (e.g., with the same hand). The one or more user input devices **113** may be configured to be operated or actuated by a thumb of a person holding the handle **110**.
[0025] In accordance with some examples of the present disclosure, the handle **110** may further include a base **114** disposed at or near a bottom of the handle **110**. In accordance with some examples, the base **114** of the handle **110** may be located closer to the body (e.g., torso) of a person holding the handle **110** than a shaft **120** of the pet toy **100**. In accordance with some examples, as described hereinafter in greater detail, a light source **140** may be disposed in the base **114** of the handle **110**. In some examples, the light source **140** disposed in the base **114** may be configured to illuminate the one or more optical strands **130**. In some examples, the light source **140** disposed in the base may be another or additional light source, which may not illuminate the one or more optical strands **130**, but instead may provide an additional source of illumination configured to attract or capture the attention of a pet. For example, the light source **140** disposed in the base **114** may be disposed within the handle **110** of the pet toy **100** and configured to illuminate or emit light through the handle **110**, so as to attract or capture the attention of a pet.
[0026] The base **114** of the handle **110** may have various shapes. For example, the base **114** may be rounded. In accordance with some examples, the base **114** may be shaped to represent one or more aspects of an object or animal. For example, as shown in FIG. **1**, the base **114** may be shaped to represent the head of a cat. In other examples, the base **114** may be shaped to represent a fruit or vegetable. The base **114** may be shaped to represent or have the shape of any item, object, or living thing.

[0027] The pet toy **100** may further include one or more optical strands **130** disposed within the shaft **120**. The one or more optical strands **130** may be configured to radiate light generated by the one or more light sources **140**. In some examples, as described hereinafter in greater detail, the at least one light source **140** may be disposed proximate to the one or more optical strands **130** and the one or more optical strands **130** may illuminate as light from the at least one light source **140** travels through and/or along the one or more optical strands **130**. In accordance with other examples, the one or more light sources **140** may be disposed along the one or more optical strands **130**.

[0028] Referring to FIG. **1**, in accordance with some examples, the one or more optical strands **130** may extend along the length of the shaft **120**. In accordance with some examples, as shown in FIG. **1**, the one or more optical strands **130** may extend along the entire length of the shaft **120** in a predetermined pattern. For example, as shown in FIG. **1**, the one or more optical strands **130** may be disposed in a repeating and continuous (e.g., linear) pattern, such as in a twisted fashion to create a wave-like pattern. A pattern of the one or more optical strands **130** may be configured to attract or capture the attention of pets, encouraging interaction with the pet toy **100**.

[0029] Referring to FIG. **3**, a partial cross section view of the pet toy **100** is illustrated in accordance with one example of the present disclosure. In accordance with some examples of the present disclosure, the optical strands **130** and at least one light source **140** included in the pet toy **100** may be the optical strands **330** and the at least one light source **340** illustrated and described below with respect to FIG. **3**. In accordance with other examples, the optical strands **130** and the at least one light source **140** included in the pet toy **100** may be the optical strands **430** and the at least one light source **440** illustrated and described below with respect to FIG. **4**.

[0030] According to some examples, as illustrated in FIG. **3**, the one or more optical strands **330** may receive light generated by the at least one light source **340** and illuminate as light from the at least one light source **340** travels along and out of the optical strands **330**. Accordingly, in some examples, the one or more optical strands **330** may be made of a clear (e.g., transparent, semitransparent) material. The one or more optical strands **330** may be made of a clear (i.e., transparent or translucent) plastic or plastics. For example, the one or more optical strands **330** may be made of polycarbonate (PC), polyethylene terephthalate (PET), polyvinyl chloride (PVC), acrylic (e.g., polymethyl methacrylate), thermoplastic polyurethane (TPU), or the like. In other examples, the one or more optical strands may be made of other materials. According to some examples, the one or more optical strands **330** may be flexible.

[0031] In some examples, as shown in FIG. **3**, the at least one light source **340** may be disposed in the handle **110** of the pet toy **100**. For example, the light source **340** may be disposed between the first panel **111** and the second panel **112** of the handle **110**. According to some examples, the at least one light source **340** may be at least one light emitting diode (LED). For example, the at least one LED may be multicolor or red, green, blue (RGB). For example, an RGB LED may be configured to emit light having a wavelength or color corresponding to the full spectrum of visible light. According to other examples, the at least one light source **340** may be at least one incandescent bulb, at least one fluorescent light, or another light source. According to some examples, the at least one light source **340** may comprise a plurality of LEDs. For example, the at least one light source may comprise two or more LEDs, each LED configured to emit light having a different wavelength or color.

[0032] In accordance with some examples of the present disclosure, the light source **340** may be disposed proximate to the one or more optical strands **330** and/or the shaft **120**. For example, the light source **340** may be disposed between the one or more optical strands **330** and/or shaft **120** and a user input device **113** included in the handle **110**. In accordance with some examples, the pet toy **100** may further include an (e.g., opaque) cover, screen, or blind **350** around a portion of any one, any combination, or all of (1) the light source **340**; (2) the one or more optical strands **330**; or (3) the shaft **120**. The blind **350** may be configured to block light emitted by the light source **340**, such that more of the light generated by the light source **340** is directed to the one or more optical strands **330** and/or the shaft **120**. According to the present disclosure, the blind **350** may be provided as a housing surrounding a portion of the light source **340**, the one or more optical strands **350**, and/or the shaft **120**. According to other examples, the blind **350** may be provided as a covering, for example, the blind **350** may be an (e.g., opaque) tape or coating (e.g., paint) applied to a portion of the light source **340**, the one or more optical strands **330**, and/or the shaft **120**.

[0033] The one or more optical strands **330**, or at least one end of each of the one or more optical strands **330**, may be disposed proximate to the at least one light source **340**. According to some examples, as shown in FIG. **3**, the one or more optical strands **330** may extend into the handle **110** of the pet toy **100**. For example, the one or more optical strands **330** may abut or contact the at least one light source **340** near the proximal end **121** of the shaft **120**. In another example, the one or more optical strands **330** may extend further into the handle **110** and abut or contact or couple to the at least one light source **340** located at or near the base **114** of the handle **110**. As noted above, the one or more optical strands **330** may be configured to radiate light generated by the at least one light source **340**. Specifically, the one or more optical strands **330** may be disposed proximate to the at least one light source **340**, such that when the at least one light source **340** generates or emits light, the light generated by the at least one light source **340** travels along or through the one or more optical strands **330**. In these examples, as light generated by the at least one light source **340** travels through the one or more optical strands **330**, the one or more optical strands **330** may illuminate, radiating the light generated by the at least one light source **340**.

[0034] According to some examples, the one or more optical strands **330** may be multicolored. In some examples, each of the one or more optical strands **330** may be configured to absorb different wavelengths (e.g., colors) of light, such that light having a different wavelength (e.g., color) exits each of the plurality of optical strands **330**. According to some examples, different portions of the same optical strand **330** may be configured to absorb different wavelengths (e.g., colors), such that different portions of the same optical strand **330** may radiate light having a different wavelength (e.g., color).

[0035] In accordance with some examples of the present disclosure, the one or more optical strands **330** may be configured to radiate light according to a plurality of (e.g., two or more) functional or operational modes. As noted above, a functional or operational mode of the at least one light source **340**, and thus, the one or more optical strands **330**, may be changed by actuating or operating the one or more user input devices **113**. In addition to changing an operational mode of the at least one light source **340**, the one or more user input devices **113** may be configured to turn the at least one light source **340** on and/or off.

[0036] The one or more optical strands **330** may be configured to radiate light according to a plurality of operational modes. The plurality of operational modes may attract or capture the attention of pets, encouraging pets to interact with the pet toy **100**. In accordance with some examples, the at least one light source **340** may facilitate or dictate an operational mode of the pet toy **100**.

[0037] The plurality of operational modes may include two or more modes in which the one or more optical strands **330** radiate light having a different wavelength or color. In some examples, the at least one light source **340** may include one or more light sources configured to emit light having various wavelengths or colors. In these examples, the one or more light sources **340** may be configured or controlled to emit light having a specific wavelength or color according to an operational mode of the pet toy **100**. For example, the at least one light source **340** may include at least one RGB LED capable of emitting light having various wavelengths or colors. In accordance with other examples, the at least one light source **340** may include a plurality of light sources configured to emit light having different wavelengths of colors and one or more of the plurality of light sources may be configured or controlled to emit light having a specific wavelength or color according to an operational mode of the pet toy **100**.

[0038] The plurality of operational modes may include one or modes in which the at least one light source **340** flashes or blinks on and off, such that the one or more optical strands **330** emit light in a flashing or blinking pattern. The at least one light source **340** may flash or blink in a regular or irregular pattern, such that light is radiated by the one or more optical strands **330** in a regular or irregular pattern. According to some examples, the plurality of operational modes may include two or more modes in which the light source **340** flashes or blinks, for example, at different speeds or intervals.
[0039] The plurality of operational modes may include one or more modes in which the one or more optical strands **330** radiate light which changes wavelengths or colors in a continuous cycle or pattern. For example, the at least one light source **340** may emit light which changes wavelengths or colors in a continuous cycle or pattern. Additionally, the one or more operational modes may include one or more modes in which the brightness of light radiated by the one or more optical strands **330** varies over time. For example, a brightness of light radiated by the one or more optical strands **330** may get brighter and dimmer in a continuous or repetitive pattern. Other operational modes that result in the light radiated by the one or more optical strands **330** changing may be used. A user may change operational modes by actuating the one or more user input devices **113**, such as by pressing a button a certain number of times, holding the button down for a predetermined period, and the like.
[0040] Referring to FIG. **4**, an optical strand **430** is illustrated in accordance with another example of the present disclosure. In accordance with some examples of the present disclosure, the one or more optical strands **130** included in the pet toy **100** may be optical strands **430** as illustrated and described below with respect to FIG. **4**. Additionally, in some examples, the at least one light source **140** included in the pet toy **100** may be the plurality of light sources **440** illustrated and described below with respect to FIG. **4**.
[0041] In some examples, as illustrated in FIG. **4**, the at least one light source **140** may comprise a plurality of light sources **440** disposed along a length of the optical strand **430**. For example, the one or more optical strands **430** may each include a base or substrate **431** and a plurality of light sources **440** may be coupled to the substrate **431**. For example, the plurality of light sources **440** may be disposed at regular or irregular intervals along the substrate **431**. According to some examples, the substrate **431** may be flexible. In some examples, the substrate **431** may be a printed circuit board (PCB). For example, the substrate **431** may be a flexible PCB including a core comprised of a flexible polymer.
[0042] According to some examples, the plurality of light sources **440** may be a plurality of LEDs. The plurality of light sources **440** may include multicolor LEDs, RGB LEDs, monochrome (e.g., single color) LEDs, or combinations thereof. According to other examples, the plurality of light sources **440** may be a plurality of incandescent bulbs, a plurality of fluorescent lights, or a plurality of other light sources. In accordance with some examples, the one or more optical strands **430** may include a trace or wire **432** connecting the plurality of light sources **440**. In some examples, the trace or wire **432** may connect the plurality of light sources **440** to a power supply and/or controller.
[0043] In accordance with some examples of the present disclosure, the one or more optical strands **430** may be configured to radiate light according to a plurality of (e.g., two or more) functional or operational modes. According to the present disclosure, the one or more optical strands **430** and the plurality of light sources **440** may be configured to radiate light according to any of the operational modes described above with respect to the optical strands **330** and the at least one light source **340** of FIG. **3**.
[0044] In addition to the operational modes described above with respect to FIG. **3**, the plurality of operational modes may include one or more operational modes in which the plurality of light sources **440** disposed along at least one of the one or more optical strands **430** sequentially emit or radiate light. For example, the plurality of lights sources **440** may turn on and off in sequence

creating a chase effect in which the plurality of light sources **440** illuminate one after another. Other operational modes configured to activate one or more of the plurality of light sources **440** in different configurations are possible.

[0045] Returning to FIG. **1**, the pet toy **100** may further include a cord **150** coupled to the shaft **120**. In some examples, as illustrated in FIG. **1**, the cord **150** may be coupled to a distal end **122** of the shaft **120**. In other examples, the cord **150** may be coupled to a center or midpoint of the shaft **120**. The cord **150** may include a proximal end disposed at or coupled to the shaft **120** and a distal end opposite the proximal end. In some examples, the cord **150** may include a loop **151**. Specifically, as illustrated in FIG. **1**, a loop **151** may be disposed at a distal end of the cord **150**. The cord **150** may be configured to be coupled to the toy assembly **160**. In some examples, the loop **151** of the cord **150** may be configured to be coupled to an elastic loop **162** of the toy assembly **160**. In some examples, the cord **150** may be made of nylon or another synthetic polymer. In other examples, the cord **150** may be made of another material.

[0046] Still referring to FIG. **1**, the pet toy **100** may further include a toy assembly **160** (i.e., toy accessory). The toy assembly **160** may include a body **161** and a loop **162**. As shown in FIG. **1**, the toy assembly **160** may be coupled to the cord **150**. For example, the loop **162** of the toy assembly **160** may be coupled to the cord **150**, for example, the loop **151** of the cord **150**. The loop **162** may be elastic and may be coupled to and/or extend from the body **161** of the toy assembly **160**. The elastic loop **162** may be configured to couple the toy assembly **160** to the cord **150**. In accordance with some examples of the present disclosure, the toy assembly **160** may be removably coupled to the cord **150**. For example, the toy assembly **160** may be removably coupled to the cord **150**, such that a person can swap out or change a toy assembly **160** coupled to the cord **150**. In one or more examples, the loop **162** of the toy assembly **160** may be removably coupled to the cord **150**. For example, the loop **162** of the toy assembly **160** may be interlocked with a loop **151** in the cord **150**. In other examples, the loop **162** and the cord **150** may be tied to one another. In other examples, the toy assembly **160** may be (e.g., removably) coupled to the cord **150** in another way.

[0047] The body **161** of the toy assembly **160** may include a case or cover **164** and an internal compartment. The cover **164** may define an outer surface of the toy assembly **160**. The cover **164** may be comprised of a plush textile. In some examples, the cover **164** may be comprised of a plush textile having pile (e.g., loops or strands) extending from the base textile. The cover **164** may be comprised of a polyester plush textile or fabric.

[0048] The internal compartment may be disposed within the cover **164** of the toy assembly **160**. In some examples, the cover **164** may define a boundary of the internal compartment. The internal compartment of the toy assembly **160** may be stuffed with a filler material. The filler material may give the body **161** a voluminous, three-dimensional shape. In some examples, the filler material may be made of polyester or another synthetic material. In some examples, the filler material may include catnip in order to attract or capture the attention of a pet. For example, the catnip may encourage a pet to bat at, grab, bite, chew or otherwise play with the toy assembly **160** hanging (e.g., via the cord **150**) from the shaft **120**.

[0049] The body **161** of the toy assembly **160** may have various shapes. In some examples, as shown in FIG. **1**, the body **161** may be shaped to represent a star. In other examples, the body **161** may be shaped to represent an animal, for example, a mouse. In still other examples, the body **161** may be configured to represent a fruit or vegetable. The body **161** may be shaped to represent or have the shape of any item, object, or living thing.

[0050] In some examples, the body **161** may further include one or more designs or images printed and/or embroidered on the cover **164** of the body **161**, some of which are discussed below. The body **161** may include one or more decorative elements **165**. The decorative element(s) **165** may be coupled to the cover **164** of the toy assembly **160**. The decorative element **165** may be, for example, a design printed or embroidered on the cover **164**. For example, a facial feature or features (e.g., eyes, nose, ears, mouth, eyebrows, or the like) may be printed or embroidered on the cover **164**. In some examples, the toy assembly **160** may further include an attractant **166**, for example, a tuft of faux fur or feather coupled to the cover **164** of the body **161**.

[0051] One or more decorative elements **165** and/or attractants **166** may be sewn, glued, or otherwise coupled to the cover **164** of the toy assembly **160**. The decorative element(s) **165** and/or attractant(s) **166** may be comprised of a different material, have a different color (or an assortment of colors), and/or have a different texture than the cover **164**. Specifically, in some examples, the cover **164** and the decorative element **165** may have different textures, so as to attract a pet. In some examples, a textile or fabric of which the cover **164** is made and a textile or fabric of which the decorative element **165** is made may include pile or strands having different lengths. In other examples, the cover **164** and the decorative element **165** or attractant **166** may have contrasting colors, so as to attract a pet.

[0052] In accordance with some examples, the toy assembly **160** may further include an additional light source **167**. For example, the toy assembly **160** may include an additional light source **167** disposed in an internal compartment of the toy assembly **160**, visible from an exterior of the toy assembly **160** (e.g., through the cover **164**) or may include an additional light source **167** coupled to an exterior surface of the cover **164**. According to some examples, the toy assembly **160** may further include an additional user input device, for example, a button, or switch, disposed in the internal compartment of the toy assembly **160**. The additional user input device may be configured to turn the additional light source **167** on and/or off and to change an operational mode of the additional light source **167**. The additional light source **167** may be configured to emit or radiate light according to any of the operational modes described herein. For example, the additional light source **167** may be configured to emit or radiate light in accordance with any of the operational modes described above with respect to the optical strands **330** or **430** described above with respect to FIGS. **3** and **4**. In some examples, the toy assembly **160** may further include a power supply and/or a controller connected to the additional light source and/or the additional user input device. The additional light source **167** may illuminate, attracting or capturing the attention of a pet. Further, the additional light source **167** may be configured to emit light according to two or more operational modes, providing a dynamic playing experience capturing a pet's attention for long or extended periods of time and/or over multiple playing sessions.

[0053] In some examples, the toy assembly **160** may further include a noisemaker. The noisemaker may be disposed within the body **161** of the toy assembly **160** (e.g., within the internal component) or coupled to the body **161** as an attractant **166**. The noisemaker may be configured to make or produce a noise as the toy assembly **160** (and thus the noisemaker) moves. The noisemaker may include any of a bell, a rattle, crinkle paper, a whistle, a clicker, an electronic noise producing device, or the like, or any combination thereof.

[0054] Referring to FIG. **5**, a block diagram of the pet toy **100** is illustrated in accordance with one example of the present disclosure. As shown in FIG. **5**, the pet toy **100** may include at least one light source **140**, at least one user input device **113**, a controller **520**, and a power supply **510**. The at least one light source **140** may be the at least one light source **340** as illustrated and described with respect to FIG. **3** or the at least one light source **440** as illustrated and described with respect to FIG. **4**.

[0055] The power supply **510** may be connected to the at least one light source **140** and/or the controller **520**. The power supply **510** may be configured to provide power (e.g., electric current) to the at least one light source **140**. The power supply **510** may be a battery pack or compartment configured to receive one or more disposable or rechargeable batteries. In some examples, as shown in FIG. **6**, the handle **110** of the pet toy may include a battery compartment **610** configured to receive one or more batteries.

[0056] In some examples, as shown in FIG. **5**, the pet toy **100** may include a controller **520**. The controller **520** may be connected to and/or in communication with the power supply **510**, the at least one light source **140**, and the at least one user input device **113**. The controller **520** may be configured to provide one or more control signals and/or electric current to the at least one light source **140**. The one or more control signals and/or electric current provided by the controller **520** may turn the at least one light source **140** on and/or off, change a color of light emitted by the at least one light source **140**, change a brightness of the at least one light source **140**, or control the at least one light source **140**, such that light is radiated by the one or more optical strands **130** in accordance with one or more operational or functional modes, for example, one or more of the operational modes described above with respect to FIGS. **2** and **3**.

[0057] The controller **520** may include a processor and memory. The memory may store one or more sets of rules or algorithms, for example, for controlling the at least one light source **140** such that light is radiated from the one or more optical strands **130** in accordance with a specific operational mode, and the processor may implement or execute the one or more sets of rules or algorithms. In accordance with some examples, the controller **520** may receive one or more user input signals from the at least one user input device **113** and control operation of the at least one light source **140** in accordance with the one or more user input signals.

[0058] In accordance with some examples, as illustrated in FIG. **7**, the pet toy **100** may include two user input devices. For example, the pet toy **100** may include a first user input device **710** and a second user input device **720**. The first user input device **710** and the second user input device **720** may each be a user input device **113** as described above with respect to FIGS. **1** and **2**. In accordance with some examples, the first user input device **710** and the second used input device **720** may be configured to perform different functions. For example, actuation of the first user input device **710** may turn the at least one light source **140** on and/or off, such that the one or more optical strands **130** are lit or illuminated (e.g., radiate light) or are unlit or unilluminated (e.g., do not radiate light), respectively. In accordance with some examples, when the first user input device **710** turns the at least one light source **140** on and/or off, actuation of the second user input device **720** may change any of a color of light emitted by the at least one light source **140**, a brightness or intensity of light emitted by the at least one light source **140**, and/or an operational or functional mode of the pet toy, for example, as described above with respect to FIGS. **3** and **4**.

[0059] In accordance with other examples, the one or more optical strands **130** may radiate light having a first color, a first intensity, or according to a first operational mode when the first user input device **710** is depressed and radiate light having a second color, a second intensity, or according to a second operational mode when the second user input device **720** is depressed.

[0060] For clarity of disclosure, the terms "proximal" and "distal" are defined herein relative to a point of attachment of an element or relative to a person using the pet toy **100** of the present disclosure. The term "proximal" refers to the position of an element closer to the point of attachment or closer to a person holding the pet toy **100**. The term "distal" refers to the position of an element further away from the point of attachment or further away from a person holding the pet toy **100**.

[0061] While the present disclosure has been described with reference to specific examples, which are intended to be illustrative only and not to be limiting of the disclosure, it should be apparent to those of ordinary skill in the art that changes, additions and/or deletions may be made to the disclosed embodiments without departing from the spirit and scope of the disclosure.

[0062] The foregoing description is given for clarity of understanding only, and no unnecessary limitations should be understood therefrom, as modifications within the scope of the disclosure may be apparent to those having ordinary skill in the art.

[0063] When a component, device, element, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element, should be considered herein as being "configured to" meet that purpose or perform that operation or function.

Claims

1. A pet toy comprising: a handle; at least one light source; a transparent or semitransparent shaft extending from the handle, the shaft including one or more optical strands configured to radiate light generated by the at least one light source; a cord attached to the shaft; and a toy assembly coupled to the cord.
2. The pet toy of claim 1, wherein the at least one light source is disposed in the handle.
3. The pet toy of claim 1, wherein the at least one light source comprises a plurality of light sources disposed along each of the one or more optical strands.
4. The pet toy of claim 1, wherein the toy assembly includes a body and a loop coupled to the body.
5. The pet toy of claim 4, wherein the body includes an additional light source.
6. The pet toy of claim 1, wherein the one or more optical strands comprise one or more light emitting diode strands and the at least one light source comprises a plurality of light emitting diodes disposed along each of the one or more light emitting diode strands.
7. The pet toy of claim 6, wherein the plurality of light emitting diodes disposed along each of the one or more light emitting diode strands includes a first light emitting diode configured to generate a first color of light and a second light emitting diode configured to generate a second color of light.
8. The pet toy of claim 1, wherein the one or more optical strands are arranged within the shaft so as to form a recurring pattern or shape.
9. The pet toy of claim 1, wherein a portion of each of the one or more optical strands are disposed in the handle.
10. The pet toy of claim 1, wherein the one or more optical strands abut the at least one light source.
11. The pet toy of claim 1, wherein the one or more optical strands are configured to radiate light generated by the at least one light source in accordance with two or more operational modes.
12. A pet toy comprising: a handle; at least one light source; a transparent shaft extending from the handle, the shaft including one or more optical strands configured to radiate light generated by the at least one light source; a cord attached to the shaft; and a toy assembly coupled to the cord, the toy assembly including at least one additional light source.
13. The pet toy of claim 12, wherein the at least one light source is disposed in the handle.
14. The pet toy of claim 12, wherein the at least one light source is disposed in the shaft.
15. The pet toy of claim 12, wherein the at least one light source comprises a plurality of light sources disposed along each of the one or more optical strands.
16. The pet toy of claim 12, wherein the handle incudes two or more user input devices.
17. A pet toy comprising: a handle including at least one user input device; at least one light source; a transparent shaft extending from the handle, the shaft including one or more optical strands configured to radiate light generated by the least one light source; a cord attached to a distal end of the shaft opposite the handle; and a toy assembly coupled to the cord.
18. The pet toy of claim 17, wherein the one or more optical strands are configured to radiate light generated by the at least one light source in accordance with two or more operational modes.
19. The pet toy of claim 18, wherein the user input device is configured to change an operational mode of the two or more operation modes of the pet toy.
20. The pet toy of claim 17, wherein the one or more optical strands are in contact with the at least one light source.